

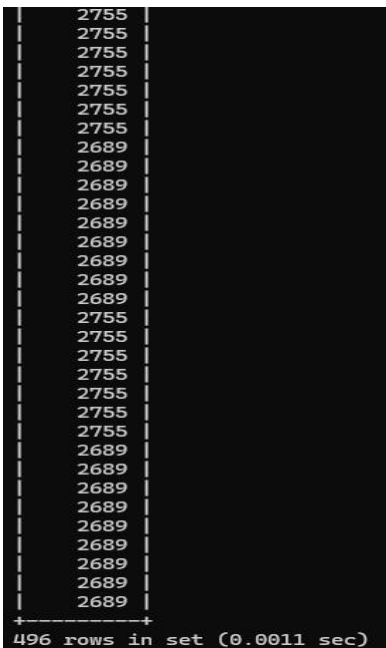
DBMS LAB4

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Aim: To implement and execute advanced SQL queries on a relational database using joins, subqueries, set operations, aggregate functions, grouping, and conditional operators for effective data retrieval and analysis.

Q1. Find match Ids of all the matches in the database in which Tendulkar batted.

```
mysql> SELECT MatchID FROM Batting, Player;
```

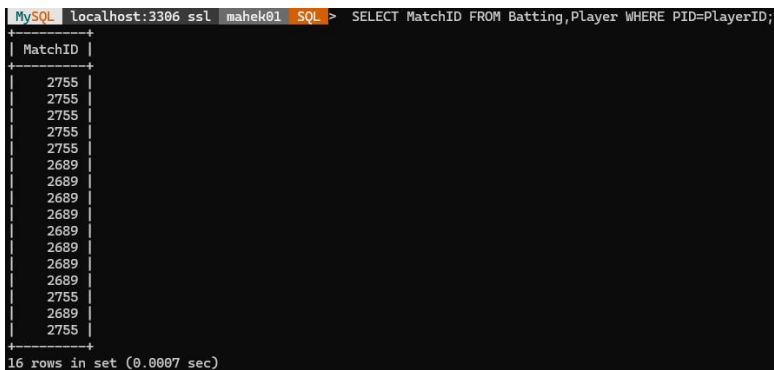


A screenshot of a MySQL terminal window. The terminal shows the output of the query 'SELECT MatchID FROM Batting, Player;'. The output is a list of MatchID values, with 2755 appearing 16 times and 2689 appearing 33 times. The terminal indicates that there are 496 rows in the set, and the query was executed in 0.0011 seconds.

MatchID
2755
2755
2755
2755
2755
2755
2755
2689
2689
2689
2689
2689
2689
2689
2689
2689
2755
2755
2755
2755
2755
2689
2689
2689
2689
2689
2689
2689
2689
2689
2689
2689
2689

496 rows in set (0.0011 sec)

```
mysql> SELECT MatchID FROM Batting,Player WHERE PID=PlayerID;
```



A screenshot of a MySQL terminal window. The terminal shows the output of the query 'SELECT MatchID FROM Batting,Player WHERE PID=PlayerID;'. The output is a list of MatchID values, with 2755 appearing 10 times and 2689 appearing 6 times. The terminal indicates that there are 16 rows in the set, and the query was executed in 0.0007 seconds.

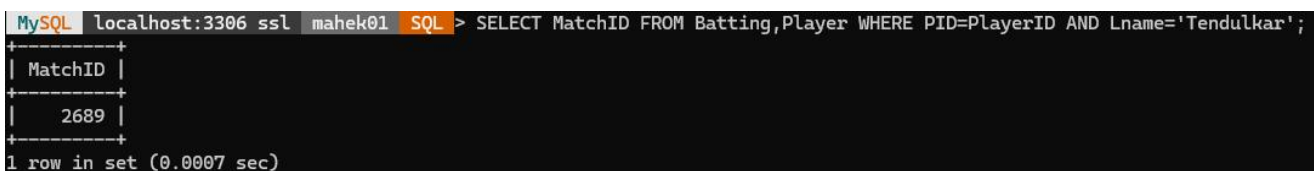
MatchID
2755
2755
2755
2755
2755
2755
2689
2689
2689
2689
2689
2689
2755
2689
2755

16 rows in set (0.0007 sec)

```
mysql> SELECT MatchID FROM Batting,Player WHERE PID=PlayerID AND  
Lname='Tendulkar';
```

OR

```
mysql> SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM  
Player WHERE LName='Tendulkar');
```



A screenshot of a MySQL terminal window. The terminal shows the output of the query 'SELECT MatchID FROM Batting,Player WHERE PID=PlayerID AND Lname='Tendulkar';'. The output is a single row with the MatchID value 2689. The terminal indicates that there is 1 row in the set, and the query was executed in 0.0007 seconds.

MatchID
2689

1 row in set (0.0007 sec)

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch,Batting,Player;
```

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch WHERE MatchID IN
(SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player
WHERE LName ='Dhoni'));
```

Using Join

Q3. Find match Ids of all matches in which Tendulkar batted.

```
MySQL localhost:3306 ssl mahek01 SQL> SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player WHERE Lname='Tendulkar');
+-----+
| MatchID |
+-----+
|      2689 |
+-----+
1 row in set (0.0008 sec)
```

Q4. Find Ids of all players that have both bowled and batted in the ODI match 2689

```
mysql> SELECT PID FROM Batting WHERE MatchID=2689 AND PID IN (SELECT PID FROM Bowling WHERE MatchID=2689);
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT PID FROM Batting WHERE MatchID=2689 AND PID IN (SELECT PID FROM Bowling WHERE MatchID=2689);
```

PID
99001
24001
23001
98002

```
4 rows in set (0.0139 sec)
```

Q5. Find Ids of all players that have either bowled or batted (or did both) in ODI match 2689

```
mysql> SELECT PID FROM Batting WHERE MatchID=2689 OR PID IN (SELECT PID FROM Bowling WHERE MatchID=2689);
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT PID FROM Batting WHERE MatchID=2689 OR PID IN (SELECT PID FROM Bowling WHERE MatchID=2689);
```

PID
23001
89001
23001
25001
99002
95001
24001
99001
27001
98002
98002

```
11 rows in set (0.0037 sec)
```

Using Union

```
mysql> SELECT PID FROM Batting WHERE MatchID=2689 UNION (SELECT PID FROM Bowling WHERE MatchID=2689);
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT PID FROM Batting WHERE MatchID=2689 UNION (SELECT PID FROM Bowling WHERE MatchID=2689);
```

PID
89001
23001
25001
99002
95001
24001
99001
27001
98002

```
9 rows in set (0.0008 sec)
```

Q6. Find Ids

```
mysql> SELECT PID FROM Batting WHERE MatchID=2689 AND PID NOT IN (SELECT PID FROM Bowling WHERE MatchID=2689);
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT PID FROM Batting WHERE MatchID=2689 AND PID NOT IN (SELECT PID FROM Bowling WHERE MatchID=2689);
```

PID
89001
25001
99002
95001
27001

```
5 rows in set (0.0007 sec)
```

Q7. Find match Ids where Tendulkar has played.

```
mysql> SELECT MatchID FROM Batting,(SELECT PlayerID FROM Player WHERE LName='Tendulkar') ST WHERE PID=ST.PlayerID;
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT MatchID FROM Batting,(SELECT PlayerID FROM Player WHERE LName='Tendulkar') ST WHERE PID=ST.PlayerID;
+-----+
| MatchID |
+-----+
|      2689 |
+-----+
1 row in set (0.0027 sec)
```

Q8. Find match Ids of all those matches in which player 27001 bats and makes more runs than he made at every match he played at Colombo

```
mysql> SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns > ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo'));
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns > ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo'));
+-----+
| MatchID |
+-----+
|      2689 |
+-----+
1 row in set (0.0028 sec)
```

Q9. Find match Ids of all those matches in which player 27001 has scored minimum runs than he made at every match he played at Brisbane

```
mysql> SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns < ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Brisbane'));
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns < ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Brisbane'));
+-----+
| MatchID |
+-----+
|      2755 |
+-----+
1 row in set (0.0028 sec)
```

Q10(a). Find Ids and scores of players who scored less than 75 but more than 50 in Colombo.

```
mysql> SELECT PID,NRuns FROM Batting WHERE NRuns BETWEEN 51 AND 74 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo');
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT PID,NRuns FROM Batting WHERE NRuns BETWEEN 51 AND 74 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo');
+-----+-----+
| PID | NRuns |
+-----+-----+
| 25001 | 71 |
| 91001 | 60 |
+-----+-----+
2 rows in set (0.0015 sec)
```

Q10(b). Find Ids and scores of players who scored greater than 49 runs in Colombo.

```
mysql> SELECT PID,NRuns FROM Batting WHERE NRuns > 49 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo');
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT PID,NRuns FROM Batting WHERE NRuns > 49 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo');
+-----+
| PID | NRuns |
+-----+
| 25001 | 71 |
| 91001 | 60 |
+-----+
2 rows in set (0.0004 sec)
```

Q11. Find the player Ids of those players whose date of first match (FTest) is not given in the database.

mysql> SELECT PlayerID FROM Player WHERE FTest IS NULL;

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT PlayerID FROM Player WHERE FTest IS NULL;
+-----+
| PlayerID |
+-----+
| 27001 |
+-----+
1 row in set (0.0005 sec)
```

Q12. Find names of the player who batted in match 2689

mysql> SELECT FName,LName FROM Player WHERE EXISTS (SELECT * FROM Batting WHERE PlayerID=PID AND MatchID=2689);

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT FName,LName FROM Player WHERE EXISTS ( SELECT * FROM Batting WHERE PlayerID=PID AND MatchID=2689);
+-----+
| FName | LName |
+-----+
| Sachin | Tendulkar |
| Yuvraj | Singh |
| MS | Dhoni |
| Adam | Gilchrist |
| Ricky | Ponting |
| Andrew | Symonds |
| Breet | Lee |
| Praveen | Kumar |
| Harbhajan | Singh |
+-----+
9 rows in set (0.0005 sec)
```

Q13. Find player Ids of players who have scored more than 30 in every ODI match that they have played.

mysql> SELECT PID AS PlayerID FROM Batting b1 WHERE NOT EXISTS (SELECT * FROM Batting b2 WHERE b1.PID=b2.PID AND NRuns < 31);

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT PID AS PlayerID FROM Batting b1 WHERE NOT EXISTS (SELECT * FROM Batting b2 WHERE b1.PID=b2.PID AND NRuns < 31);
+-----+
| PlayerID |
+-----+
| 24001 |
| 25001 |
| 25001 |
| 89001 |
| 91001 |
+-----+
5 rows in set (0.0007 sec)
```

Q14. Find the names of players that have batted in all the ODI matches in Melbourne that are in the database.

SELECT FName, LName FROM Player WHERE NOT EXISTS (SELECT * FROM ODIMatch WHERE Ground = 'Melbourne' AND NOT EXISTS (SELECT * FROM Batting WHERE Batting.PID = Player.PlayerID AND Batting.MatchID = ODIMatch.MatchID));

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT FName, LName FROM Player WHERE NOT EXISTS (SELECT * FROM ODIMatch WHERE Ground = 'Melbourne' AND NOT EXISTS (SELECT * FROM Batting WHERE Batting.PID = Player.PlayerID AND Batting.MatchID = ODIMatch.MatchID));
Empty set (0.0007 sec)
```

Q15. Find the number of players that bowled in ODI match 2689

mysql> SELECT COUNT(*) AS NBowlers FROM Bowling WHERE MatchID=2689;

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT COUNT(*) AS NBowlers FROM Bowling WHERE MatchID=2689;
```

NBowlers
4

1 row in set (0.0042 sec)

Q16. Find the average batting score of all the players that batted in the ODI Match 2689

```
mysql> SELECT AVG(NRuns) AS AvgRuns_2689 FROM Batting WHERE MatchID=2689;
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT AVG(NRuns) AS AvgRuns_2689 FROM Batting WHERE MatchID=2689;
```

AvgRuns_2689
25.2222

1 row in set (0.0031 sec)

Q17. Find the youngest player in the database

```
mysql> SELECT FName AS Youngest_Player FROM Player WHERE YBorn=(SELECT MAX(YBorn) FROM Player);
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT FName AS Youngest_Player FROM Player WHERE YBorn=(SELECT MAX(YBorn) FROM Player);
```

Youngest_Player
Ishant

1 row in set (0.0031 sec)

Q18. Find the number of players in the database from each country

```
mysql> SELECT Country, COUNT(*) AS NPlayers FROM Player GROUP BY Country;
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT Country, COUNT(*) AS NPlayers FROM Player GROUP BY Country;
```

Country	NPlayers
India	10
Australia	5
West Indies	1
Sri Lanka	3
Pakistan	2
Zimbabwe	1
South Africa	4
New Zealand	2
England	1

9 rows in set (0.0007 sec)

Q19. Find batting average of each player in the database. Present the PID of each Player.

```
mysql> SELECT PID,AVG(NRuns) AS AVGR FROM Batting GROUP BY PID;
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT PID,AVG(NRuns) AS AVGR FROM Batting GROUP BY PID;
```

PID	AVGR
23001	19.0000
24001	42.0000
25001	53.5000
27001	4.5000
89001	91.0000
91001	60.0000
92002	1.0000
94002	17.0000
95001	1.0000
98002	2.5000
99001	7.0000
99002	2.0000

12 rows in set (0.0006 sec)

Solve the following

Q20. Find the batting average of each player from India in the database. Present first name of each player.

```
mysql> SELECT P.FName, SUM(B.NRuns) / COUNT(B.MatchID) AS Batting_Average
FROM Player P JOIN Batting B ON P.PlayerID = B.PID WHERE P.Country = 'India'
GROUP BY P.PlayerID, P.FName;
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT P.FName, SUM(B.NRuns) / COUNT(B.MatchID) AS Batting_Average FROM Player P JOIN Batting B ON P.PlayerID = B.PID WHERE P.Country = 'India' GROUP BY P.PlayerID, P.FName;
```

FName	Batting_Average
Yuvraj	19.0000
MS	53.5000
Praveen	4.5000
Sachin	91.0000
Harbhajan	2.5000

5 rows in set (0.0029 sec)

Q21. Find the average score for each player when playing against Australia

```
SELECT P.FName, AVG(B.NRuns) AS Average_Score FROM Player P JOIN Batting B ON
P.PlayerID = B.PID JOIN ODIMatch M ON B.MatchID = M.MatchID WHERE M.Team1 =
'Australia' OR M.Team2 = 'Australia' GROUP BY P.PlayerID, P.FName;
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT P.FName, AVG(B.NRuns) AS Average_Score FROM Player P JOIN Batting B ON P.PlayerID = B.PID JOIN ODIMatch M ON B.MatchID = M.MatchID WHERE M.Team1 = 'Australia' OR M.Team2 = 'Australia' GROUP BY P.PlayerID, P.FName;
```

FName	Average_Score
Sachin	91.0000
Yuvraj	38.0000
MS	36.0000
Adam	2.0000
Ricky	1.0000
Andrew	42.0000
Breet	7.0000
Praveen	7.0000
Harbhajan	3.0000

9 rows in set (0.0080 sec)

Q22. Find the ID of players that had a higher average score for all players when they played against Sri Lanka.

```
SELECT B.PID FROM Batting B JOIN ODIMatch M ON B.MatchID = M.MatchID
WHERE M.Team1 = 'Sri Lanka' OR M.Team2 = 'Sri Lanka' GROUP BY B.PID HAVING
AVG(B.NRuns) = (SELECT MAX(AvgScore) FROM (SELECT AVG(B2.NRuns) AS
AvgScore FROM Batting B2 JOIN ODIMatch M2 ON B2.MatchID = M2.MatchID WHERE
M2.Team1 = 'Sri Lanka' OR M2.Team2 = 'Sri Lanka' GROUP BY B2.PID) AS Averages);
```

```
MySQL localhost:3306 ssl mahesh01 SQL > SELECT B.PID FROM Batting B JOIN ODIMatch M ON B.MatchID = M.MatchID WHERE M.Team1 = 'Sri Lanka' OR M.Team2 = 'Sri Lanka' GROUP BY B.PID HAVING AVG(B.NRuns) = (SELECT MAX(AvgScore) FROM (SELECT AVG(B2.NRuns) AS AvgScore FROM Batting B2 JOIN ODIMatch M2 ON B2.MatchID = M2.MatchID WHERE M2.Team1 = 'Sri Lanka' OR M2.Team2 = 'Sri Lanka' GROUP BY B2.PID) AS Averages);
```

PID
25901

1 row in set (0.0033 sec)